

1-task

```
0:00 systemd-userwork: waiting...
[akbaralievich@instance1 ~]$ ps aux | awk '$8 == "Z" {print
} '
[akbaralievich@instance1 ~]$ ps aux | awk '$8 ~ /^Z/ {print}
'
[akbaralievich@instance1 ~]$
```

```
[akbaralievich@instance1 ~]$ ps aux | awk '$8 ~ /^S/ {print}
'
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START
TIME COMMAND
root           1  0.0  1.0 49412 40960 ?        Ss   Nov13
1:29 /usr/lib/systemd/systemd --switched-root --system --de
serialize=49
root           2  0.0  0.0     0     0 ?        S    Nov13
0:00 [kthreadd]
root           3  0.0  0.0     0     0 ?        S    Nov13
0:00 [pool_workqueue_release]
root          17  0.0  0.0     0     0 ?        S    Nov13
0:01 [ksoftirqd/0]
root           19  0.0  0.0     0     0 ?        S    Nov13
0:00 [rcu_exp_par_gp_kthread_worker/0]
root           20  0.0  0.0     0     0 ?        S    Nov13
0:00 [rcu_exp_gp_kthread_worker]
root           21  0.0  0.0     0     0 ?        S    Nov13
0:03 [migration/0]
root           22  0.0  0.0     0     0 ?        S    Nov13
0:00 [idle_inject/0]
root           23  0.0  0.0     0     0 ?        S    Nov13
0:00 [cpuhp/0]
root           24  0.0  0.0     0     0 ?        S    Nov13
0:00 [cpuhp/1]
root           25  0.0  0.0     0     0 ?        S    Nov13
0:00 [idle_inject/1]
```

2-task

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	49412	41020	10636	S	0.3	1.1	0:21.21	systemd
18	root	20	0	0	0	0	I	0.3	0.0	0:05.82	rcu_preempt
835	dbus	20	0	5064	3056	2328	S	0.3	0.1	0:02.84	dbus-broker
843	root	20	0	18812	10228	7772	S	0.3	0.3	0:01.63	systemd-logind
2	root	20	0	0	0	0	S	0.0	0.0	0:00.08	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
17	root	20	0	0	0	0	S	0.0	0.0	0:00.56	ksoftirqd/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.01	rcu_exp_gp_kthread_worker
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.53	migration/0
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
25	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
26	root	rt	0	0	0	0	S	0.0	0.0	0:00.81	migration/1
27	root	20	0	0	0	0	S	0.0	0.0	0:00.57	ksoftirqd/1
29	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-events_highpri
31	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
32	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_frag_wq
33	root	20	0	0	0	0	S	0.0	0.0	0:01.52	kauditd

3-task

```

0.00 grep --color=auto firefox
[akbaralievich@instance1 ~]$ pgrep chrome
[akbaralievich@instance1 ~]$ pgrep firefox
[akbaralievich@instance1 ~]$ █

Try 'kill -help' for more information.
[akbaralievich@instance1 ~]$ pkill firefox
[akbaralievich@instance1 ~]$ pkill chrome

```

4-task

```

[0]- Terminated yes > /dev/null
[akbaralievich@instance1 Internship_Tasks]$ ps -C yes -o pid,ni,cmd
  PID  NI  CMD
  77181  0  yes
  77214  0  yes
[2]- Terminated yes > /dev/null
[akbaralievich@instance1 Internship_Tasks]$ renice -n 5 -p 77214
77214 (process ID) old priority 0, new priority 5
[akbaralievich@instance1 Internship_Tasks]$ █

```

MIB Swap: 0.0 total, 0.0 free, 0.0 used. 3099.1 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
77181	akbaral+	20	0	5400	1880	1752	R	94.4	0.1	5:57.34	yes
77214	akbaral+	25	5	5400	1872	1744	R	93.4	0.1	3:48.03	yes
1678	root	20	0	8760	6408	5220	S	4.3	0.2	0:43.83	sshd
1008	root	20	0	1836948	28904	17784	S	0.3	0.8	2:05.70	google_guest_ag
77313	akbaral+	20	0	10116	5416	3280	R	0.3	0.1	0:00.14	top
1	root	20	0	49412	41020	10636	S	0.0	1.1	0:21.26	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.08	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
17	root	20	0	0	0	0	S	0.0	0.0	0:00.57	ksoftirqd/0
18	root	20	0	0	0	0	I	0.0	0.0	0:05.83	rcu_preempt
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.01	rcu_exp_gp_kthread_worker
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.53	migration/0
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
25	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
26	root	rt	0	0	0	0	S	0.0	0.0	0:00.81	migration/1
27	root	20	0	0	0	0	S	0.0	0.0	0:00.58	ksoftirqd/1

5-task

```
[akbaralievich@instance1 proc]$ sleep 1000 &
[3] 31185
[akbaralievich@instance1 proc]$ cd 31185
[akbaralievich@instance1 31185]$ cat cmdline
sleep1000[akbaralievich@instance1 31185]$
```

```
-bash: cd: cmdline: Not a directory
[akbaralievich@instance1 30747]$ sudo cat cmdline
sleep1000[akbaralievich@instance1 30747]$ cat status
Name: sleep
Umask: 0022
State: S (sleeping)
Tgid: 30747
Ngid: 0
Pid: 30747
PPid: 28945
TracerPid: 0
Uid: 1000 1000 1000 1000
Gid: 1005 1005 1005 1005
FDSize: 256
Groups: 4 39 1000 1001 1002 1003 1004 1005
NSTgid: 30747
NSpid: 30747
NSpgid: 30747
NSSid: 28945
Kthread: 0
VmPeak: 5400 kB
VmSize: 5400 kB
VmLok: 0 kB
VmPin: 0 kB
VmHWM: 1992 kB
VmRSS: 1992 kB
RssAnon: 120 kB
RssFile: 1872 kB
RssShmem: 0 kB
VmData: 204 kB
VmStk: 132 kB
VmExe: 16 kB
VmLib: 1500 kB
VmPTE: 48 kB
VmSwap: 0 kB
HugetlbPages: 0 kB
CoreDumping: 0
```

```

[akbaralievich@instance1 30747]$ cd root
[akbaralievich@instance1 root]$ ls
afs bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys usr var
[akbaralievich@instance1 root]$ cd ..
[akbaralievich@instance1 30747]$ cd root
[akbaralievich@instance1 root]$ cd proc/
[akbaralievich@instance1 proc]$ ls
1 18 27 30412 36 441 604 726 842 bus fs loadavg self vmallocinfo
10 19 28254 30476 37 445 605 734 843 cgroups interrupts locks slabinfo vmstat
1008 2 28786 30615 38 446 606 76 856 cmdline iomem mdstat softirqs zoneinfo
1014 20 28788 30744 39 45 607 77 88 consoles ioports meminfo stat
1018 21 28935 30747 4 46 608 78 930 cpuinfo irq misc swaps
1024 22 28944 30836 40 47 609 79 94 crypto kallsyms modules sys
13 23 28945 30837 41 48 610 800 943 devices kcore mounts sysrq-trigger
14 2341 29 30843 42 49 611 81 967 diskstats keys mtrr sysvipc
1439 2350 29860 30862 43 5 612 82 969 dma key-users net thread-self
15 2357 3 31 438 50 613 828 972 driver kmsg pagetypeinfo timer_list
16 24 30025 32 439 52 686 83 acpi dynamic_debug kpagecgroup partitions tty
1678 25 30342 33 44 53 7 835 bootconfig execdomains kpagecount schedstat uptime
17 26 30410 34 440 6 71 839 buddyinfo filesystems kpageflags scsi version
[akbaralievich@instance1 proc]$

```

6-task

```

[akbaralievich@instance1 ~]$ yes > /dev/null

^Z
[1]+  Stopped                  yes > /dev/null
[akbaralievich@instance1 ~]$ jobs -l
[1]+ 31695 Stopped              yes > /dev/null
[akbaralievich@instance1 ~]$ fg %1
yes > /dev/null

^Z
[1]+  Stopped                  yes > /dev/null
[akbaralievich@instance1 ~]$ bg %1
[1]+ yes > /dev/null &
[akbaralievich@instance1 ~]$ jobs -l
[1]+ 31695 Running              yes > /dev/null &
[akbaralievich@instance1 ~]$

```

7-task



```
#!/bin/bash

cleanup() {
    echo "Caught Ctrl+C (SIGINT), performing cleanup..."
    # Add your cleanup commands here
    echo "Cleanup finished."
    exit 0
}

# Trap the SIGINT signal
trap 'cleanup' SIGINT

echo "Process running. Press Ctrl+C to stop gracefully."
# Example long-running task
while true; do
    sleep 1
done

"long_process.sh" 17L, 327B
```

17,4

All

```
[akbaralievich@instance1 ~]$ vim long_process.sh
[akbaralievich@instance1 ~]$ chmod +x long_process.sh
[akbaralievich@instance1 ~]$ ./long_process.sh
Process running. Press Ctrl+C to stop gracefully.
^CCaught Ctrl+C (SIGINT), performing cleanup...
Cleanup finished.
[akbaralievich@instance1 ~]$
```